Project Proposal - Group 4

R.A.G. Stevenaar (s2633582), R.L.M. Mulder (s2587572), R.M. Hezeman (s3450902), T.P. Angevare (s2744007) June 18, 2026

Adversarial Attacks on Class-Imbalanced Credit Risk Models

Credit risk assessment allows lenders to estimate whether a borrower will default and is therefore central to lending decisions, loss mitigation and risk-based pricing. While Machine Learning (ML) and Deep Learning (DL) models can improve predictive performance in credit data, they also increase opacity and model-risk concerns [1]. A specific threat comes from adversarial examples: small, deliberate feature changes that steer a model toward a desired target label [2]. In credit scoring, the attacker can be framed as the loan applicant, who may manipulate self-reported features within plausible bounds to obtain a favourable non-default classification. This project examines whether extreme class imbalance, where defaults are rare, makes credit risk models more vulnerable to such attacks. It compares common credit risk models under realistic constraints and evaluates feature susceptibility, training strategies, defences, robustness and governance.

Potential Datasets

We consider two publicly available credit-risk datasets: the **LendingClub** loan default dataset and the **Give Me Some Credit (GMSC)** dataset. For LendingClub, we use the binary final-outcome setup where *Fully Paid* loans are labelled as non-default and *Charged Off* loans as default. This results in 1,076,751 non-default observations and 268,559 default observations, corresponding to an imbalance ratio of approximately 4.01:1. For GMSC, the target variable *SeriousDlqin2yrs* indicates whether a borrower experienced serious financial distress within two years. This dataset contains 139,974 non-default observations and 10,026 default-like observations, resulting in a stronger imbalance ratio of approximately 13.96:1.

Dataset	Non-default	Default	Default %	Ratio
LendingClub	1,076,751	268,559	19.96%	4.01:1
GMSC	139,974	10,026	6.68%	13.96:1

LendingClub is suitable as the primary dataset because of its large scale and rich loan-level feature set, including loan amount, annual income, debt-to-income ratio, employment length, loan purpose and loan status [3]. GMSC is useful as a option because it is smaller, cleaner, and more strongly imbalanced, making it suitable for testing whether the effect of class imbalance on adversarial vulnerability also holds in a simpler credit-scoring setting [4].

Experimental Framework

We will use a three-layer methodology that combines post-positivism, quasi-experimentation, and CRISP-ML(Q). A post-positivist perspective fits this project because machine learning results are inherently probabilistic rather than deterministic. Outcomes can vary because of random seeds, hyperparameters, train-test splits, and attack settings [5]. Therefore, conclusions will be based on repeated experiments, confidence intervals, and statistical tests.

Since borrowers cannot be randomly assigned to different imbalance conditions, we will use a quasi-experimental setup. The imbalance ratio will be changed in a controlled environment, while confounders such as the pre-processing pipeline, train-test splits, and random seeds will be controlled across conditions. This allows us to study the effect of class imbalance without full random assignment [6]. The factorial design will compare imbalance level, model type, training strategy, and defence mechanism. The main dependent variables will be Attack Success Rate (ASR), minimum perturbation margin, per-feature exploitation frequency, and clean predictive performance. Validity threats such as data leakage, hyperparameter instability, and attack-budget sensitivity will be explicitly identified and mitigated.

We will use CRISP-ML(Q) as the project lifecycle instead of CRISP-DM, because it includes quality assurance, risk documentation, and monitoring for machine learning projects [7], [8]. The quality layer connects with the quasi-experimental focus on validity threats and the post-positivist view that results contain uncertainty. In the business and data understanding phase, we will define the threat model and label features based on mutability, monotonicity direction, and feasibility bounds. In data preparation, we will clean and encode the data and construct natural and controlled imbalance ratios. In modelling, logistic regression, gradient boosting, and neural networks will be trained using standard learning, class weighting, and resampling strategies. In evaluation, clean performance will be compared with constraint-aware attack success and defence performance under identical attack conditions. The monitoring and maintenance phase will translate findings into recommendations for more robust and better-governed credit risk systems.

AI statement

For this assignment, AI was used for language support, such as checking grammar, spelling, and sentence clarity. In addition, AI was used to brainstorm possible approaches for tackling the assignment. The ideas, arguments, structure, analysis, and final content of this report remain entirely our own.

References

[1] I. Mapfumo and T. Shongwe, "Performance evaluation of machine learning and deep learning models for credit risk prediction," *Journal of Risk and Financial Management*, vol. 19, no. 3, p. 210, 2026. DOI: 10.3390/jrfm19030210.

[2] S. Ghamizi et al., "Search-based adversarial testing and improvement of constrained credit scoring systems," in *Proceedings of the 28th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE '20)*, Virtual Event, USA, Nov. 2020, pp. 1089–1100. DOI: 10.1145/3368089.3409739.

[3] LendingClub, *Lending club loan data*, Kaggle dataset, 2019. [Online]. Available: https://www.kaggle.com/datasets/wordsforthewise/lending-club.

[4] Kaggle, *Give me some credit*, Kaggle competition dataset, 2011. [Online]. Available: https://www.kaggle.com/c/GiveMeSomeCredit.

[5] J. W. Creswell, *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*, 3rd ed. Thousand Oaks, CA, USA: Sage Publications, 2009.

[6] W. R. Shadish, T. D. Cook, and D. T. Campbell, *Experimental and Quasi-Experimental Designs for Generalized Causal Inference*. Boston, MA, USA: Houghton Mifflin, 2002.

[7] P. Chapman et al., "Crisp-dm 1.0: Step-by-step data mining guide," SPSS Inc., Tech. Rep., 2000.

[8] S. Studer et al., "Towards crisp-ml(q): A machine learning process model with quality assurance methodology," *Machine Learning and Knowledge Extraction*, vol. 3, no. 2, pp. 392–413, 2021.